

Connor Lane

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Education

University of California, Santa Barbara, Ph.D. in Mathematics Sept 2025 – TBD

- Advised by Francesc Castella

Rose-Hulman Institute of Technology, BS in Mathematics Summa Cum Laude Sept 2022 – May 2025

- Minors in Computer Science and Theoretical Physics
- Clarence P. Sousley Award for excellence in mathematics graduates
- Senior Thesis: “Cyclotomic Units, Volkenborn Integration, and p-adic L-functions.” Advised by Dr. Tim All

Publications & Preprints

Dispersive estimates for Dirac Operators in dimension four with obstructions at threshold energies Oct 2025

Green, W.R., Lane, C. & Lyons, B. Dispersive estimates for Dirac Operators in dimension four with obstructions at threshold energies. *Calc. Var.* 64, 292 (2025). <https://doi.org/10.1007/s00526-025-03163-7>

AGM Aquariums and Elliptic Curves Over Arbitrary Finite Fields Apr 2025

June Kayath, Connor Lane, Ben Neifeld, Tianyu Ni, Hui Xue, AGM Aquariums and Elliptic Curves Over Arbitrary Finite Fields, *Res. Number Theory* 11 (2025), no. 2, Paper No. 48, 31 pp.; MR4890995

The massless Dirac equation in three dimensions: dispersive estimates and zero energy obstructions. Jan 2025

William R. Green, Connor Lane, Benjamin Lyons, Shyam Ravishankar, Aden Shaw. The massless Dirac equation in three dimensions: Dispersive estimates and zero energy obstructions, *J. Differential Equations* 416 (2025), 449–490; MR4806982

Subspaces Spanned by Eigenforms with Nonvanishing Central L-Values Jul 2024

June Kayath, Connor Lane, Ben Neifeld, Tianyu Ni, Hui Xue. "Subspaces Spanned by Eigenforms with Nonvanishing Central L-Values" arXiv preprint arXiv:2407.00532. (2024)

Asymptotic Distribution of Residues in Pascal’s Triangle mod p Sep 2023

Connor Lane. Asymptotic Distribution of Residues in Pascal’s Triangle mod p . arXiv preprint arXiv:2309.12942 (2023)

Remote Work: Fad or Future Apr 2022

Aruni Chenxi, Nika Chuzhoy, Max Hartman, Connor Lane, and Nathan Liu "Remote Work: Fad or Future." Mathworks Math Modeling Challenge Runner-Up, <https://m3challenge.siam.org/resources/archives/2022-year-at-a-glance/2022-winningsolutions/> (2022)

Talks Given

“Math Until We Die”, Ross Mathematics Program Dorm Lecture, July 11 2026

“How To Be a Millionaire”, Ross Mathematics Program Dorm Lecture, June 25 2026

“AGM Aquariums and Elliptic Curves Over Finite Fields,” Southern California Number Theory Day, Oct 18 2025

“The Prime At Infinity and Other Lies in Algebraic Number Theory,” Ross Mathematics Program Dorm Lecture, July 24 2025

“Math Until We Die,” Ross Mathematics Program Dorm Lecture, July 7 2025

“How I Rigorously Broke a Painting,” Ross Mathematics Program Dorm Lecture, June 20 2025

“AGM Aquariums and Elliptic Curves Over Finite Fields,” Rose-Hulman Undergraduate Mathematics Conference 2025, March 29 2025

“AGM Aquariums and Elliptic Curves Over Finite Fields,” Joint Mathematics Meetings PME Undergraduate Poster Session, January 9 2025

“AGM Aquariums and Elliptic Curves Over Finite Fields,” Clemson University Undergraduate Research Symposium, July 26 2024

“Cyclic Galois Extensions and Groups on Conics,” Rose-Hulman Undergraduate Mathematics Conference 2024, April 27 2024

“Asymptotic Distribution of Residues in Pascal’s Triangle mod p ,” Joint Mathematics Meetings PME Undergraduate Poster Session, January 6 2024

“Infinite Galois Theory,” Ross Mathematics Program Dorm Lecture, July 15 2023

“The Riemann Zeta Function and a Problem on my Shirt,” Ross Mathematics Program Dorm Lecture, July 6 2023

“Pascal’s Triangle mod p ,” Rose-Hulman Undergraduate Mathematics Conference 2023, April 21 2023

Teaching Experience

Mentor, Ross Online Sep 2026 – Dec 2026

- Taught and designed a course on Quadratic Forms
- Managed an online classroom with 20+ students

Teaching Assistant, University of California, Santa Barbara Oct 2025 – Ongoing

- Ran recitation sections for undergraduate classes
- Made original classroom materials

Counselor, Ross Mathematics Program Summer 2023, 2025, 2026

- Taught and graded number theory for a “family” of 5-6 high school students
- Instructor and course designer for Algebraic Number Theory

Grader, Rose-Hulman Institute of Technology Dec 2023 – Mar 2025

- Graded for upper level classes – Real Analysis, Algebraic Geometry, Galois Theory

Tutor, Rose-Hulman Institute of Technology Sep 2023 – Mar 2025

- Tutored nearly every class offered by the math department
- Designed finals review packet for introduction to proofs

Academic Honors

Graduate Research Fellowship, National Science Foundation 2026

Clarence P. Sousley Award, Rose-Hulman Institute of Technology Math Dept. 2025

Ying Award for Best Student-Faculty Collaboration, Rose-Hulman Institute of Technology 2025

Goldwater Scholarship Winner, Goldwater Foundation 2024

Theodore Paine Palmer Award, Rose-Hulman Institute of Technology Math Dept. 2023

Putnam Top 200, Mathematical Association of America 2023-2025

Dean’s List, Rose-Hulman Institute of Technology 2022-2025

M3 Challenge Runner-Up, Society for Industrial and Applied Math 2022

Organizational Work

Organizer UCSB Directed Reading Program	2026-2027
Organizer UCSB Grad Student Gestalt Reading Group	Fall 2026
Organizer UCSB Grad Student Infinity Categories Seminar	Spring 2025

Selected Coursework

Topics in Geometry	Fall 2025
University of California, Santa Barbara: Cheeger-Gromoll splitting theorem, Cheeger-Gromoll soul theorem	
Complex Manifolds	Spring 2025
Rose-Hulman Institute of Technology: Complex analysis in several variables, Hartog's phenomenon, geometry of complex manifolds	
Forcing	Spring 2025
Rose-Hulman Institute of Technology: Independence of CH, Whiteheads problem, applications to analysis	
Programming Language Paradigms	Spring 2025
Rose-Hulman Institute of Technology: Functional and declarative programming, uncommon language features	
Topics in Algebraic Geometry	Winter-Spring 2025
Rose-Hulman Institute of Technology: Sheaf cohomology on sites, étale cohomology of schemes, Poincare duality on varieties, applications to algebraic number theory	
Advanced Elliptic Curves	Fall 2024
Rose-Hulman Institute of Technology: Integral points, non-triviality of the Tate-Shafarovich group, elliptic surfaces, Néron models	
General Relativity	Spring 2024
Rose-Hulman Institute of Technology: Einstein's field equation, black holes, applications to cosmology	
Cohomology of Number Fields	Winter 2024
Rose-Hulman Institute of Technology: The local-to-global principle, Grunwald-Wang, genus theory	
Differential Forms in Algebraic Topology	Winter 2024
Rose-Hulman Institute of Technology: De Rham cohomology, Characteristic classes, spectral sequences	